

Answer Key — Biology Grade 11

Sub-Strand 3.1: Reproduction in Animals • Lesson 2: Built for Reproduction: Dissecting Mammalian Reproductive Structures

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
A	C	D	D	B	C	D	A

USE AFTER: PREDICT PHASE — Predict: sketch of what is inside a female and male mammal (Friesian cow and frog pond images)

Q1. A Friesian cow in Kericho carries her calf for about nine months before birth. In which organ does the calf develop during this time?

- A. The uterus**
- B. The ovary
- C. The oviduct
- D. The stomach

Correct answer: A. The uterus has a thick muscular wall and a rich lining that supports the embryo as it grows. The stomach is part of the digestive system, not the reproductive system. The oviduct carries the egg, and the ovary makes eggs.

USE AFTER: OBSERVE PHASE — Part A: dissection of a small mammal (male specimen groups locating the testes and epididymis)

Q2. In a male rat, the testes hang outside the main body cavity. What is the main advantage of this position?

- A. It stores the embryo safely until birth
- B. It lets urine and sperm leave through separate tubes
- C. It keeps the sperm cooler than the body temperature**
- D. It keeps the sperm warm so they develop faster

Correct answer: C. Sperm production works best at a temperature lower than normal body temperature, and the position of the testes outside the body provides this.

USE AFTER: OBSERVE PHASE — Part A: dissection of a small mammal (male specimen: "Where does the sperm duct meet the urethra?")

Q3. A group dissecting a male rabbit finds a long, tightly coiled tube lying against the testis. What is its function?

- A. It carries urine out of the body
- B. It produces testosterone only
- C. It is where fertilisation takes place
- D. It stores sperm until they are released**

Correct answer: D. This tube is the epididymis. Its coiled structure gives a long space to store sperm. Fertilisation happens in the female oviduct, and urine leaves through the urethra.

USE AFTER: OBSERVE PHASE — Part A: dissection of a small mammal (male specimen: 'Where does the sperm duct meet the urethra?')

Q4. In a male mammal, the sperm duct joins another tube that leads through the penis. Which tube is this, and what does it carry?

- A. The epididymis, which carries urine only
- B. The sperm duct, which carries urine only
- C. The oviduct, which carries sperm only
- D. The urethra, which carries both urine and sperm**

Correct answer: D. The sperm duct joins the urethra. The urethra carries urine and sperm out of the body, though not at the same time. Some learners wrongly think urine and sperm use separate tubes all the way.

USE AFTER: OBSERVE PHASE — Part A: dissection of a small mammal (female specimen groups locating the ovaries, oviducts, uterus and cervix)

Q5. A female rabbit specimen has a pair of thin tubes lined with tiny hair-like cilia that lead from the ovaries to the uterus. What is the role of these tubes?

- A. They seal the uterus to protect the embryo
- B. They move the egg along and are where fertilisation happens**
- C. They store the calf or pups until birth
- D. They make the hormones FSH and LH

Correct answer: B. These are the oviducts. Their cilia move the egg towards the uterus, and fertilisation takes place in them. The cervix is the muscular seal, and FSH and LH come from the pituitary gland.

USE AFTER: EXPLAIN PHASE — Structure-function table and Think-Pair-Share (explaining how one organ's structure helps it do its job)

Q6. Which feature of the uterus makes it well adapted to support a developing embryo?

- A. It has a thin wall so that the embryo can leave quickly
- B. It makes the eggs that will be fertilised
- C. It has a thick muscular wall and a rich lining where the embryo implants**
- D. It has cilia that carry the sperm to the egg

Correct answer: C. The rich lining allows the embryo to implant and receive support, and the muscular wall can stretch and protect the growing embryo. Eggs are made in the ovaries, and cilia are found in the oviducts.

USE AFTER: EXPLAIN PHASE — Structured debate: 'producing many gametes (frog) or protecting a single embryo inside the mother (cow/human)?'

Q7. A frog near Kisumu releases thousands of eggs into the pond, but a cow usually produces one calf at a time. Which statement best explains this difference?

- A. Cows have no ovaries, so they can only make one egg in a lifetime
- B. A cow's embryo grows in the stomach, which only has room for one
- C. Frogs have a uterus that can hold thousands of embryos
- D. Frog eggs are fertilised outside the body and many are lost, while the cow's single embryo is protected and nourished inside the mother**

Correct answer: D. External fertilisation exposes eggs to loss and predators, so frogs depend on large numbers. Mammals invest in protecting one embryo inside the uterus. Cows do have ovaries, and frogs do not develop embryos in a uterus.

Q8. A woman's oviducts are blocked, so the egg and sperm cannot meet. Which statement about how IVF can help is correct?

- A. Fertilisation can happen in a dish, and the early embryo is then placed in the uterus to develop**
- B. The embryo is placed in the stomach, where it grows
- C. The baby grows wholly in a test tube until it is born
- D. The sperm are made in the uterus instead of the testes

Correct answer: A. IVF bypasses the blocked oviduct by letting fertilisation happen outside the body. The embryo still develops in the mother's uterus. The belief that IVF babies grow wholly in a test tube is a common misconception.